Quiz Q1. Interpret Example 2's 3×2 matrix $\mathbf{A}$ as user-item data: three users and two items. Which statement correctly interprets

$$\mathbf{A}\mathbf{v}_i \;=\; \sigma_i \mathbf{u}_i \,?$$

A. The matrix $\mathbf{A}$ converts an item-space input vector into a user-space output vector.

B. The matrix $\mathbf{A}$ converts one user's latent embedding into one item's latent embedding.

C. The matrix $\mathbf{A}$ converts a user-space input vector into an item-space output vector.

D. The matrix $\mathbf{A}$ converts one item's latent embedding into one user's latent embedding.

Quiz Q1 answer. The correct answer is A.

The input $\mathbf{v}_i \in \mathbb{R}^2$ has one entry per item. Multiplication by $\mathbf{A} \in \mathbb{R}^{3\times 2}$ produces an output in $\mathbb{R}^3$, with one entry per user.

Recall Example 2: $\sigma_1 = 30$, $\mathbf{v}_1 = (3,4)^\mathsf{T}/5$ and $\mathbf{u}_1 = (2,2,1)^\mathsf{T}/3$. Thus

$$\begin{pmatrix} 20 & 10 \\ 8 & 19 \\ -2 & 14 \end{pmatrix} \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \\ 10 \end{pmatrix} = 30 \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

This maps a pattern across items to a corresponding pattern across users. The singular vectors are columns, not the individual user or item embeddings stored in rows.

Quiz Q2. The validation pairs have ratings $r_{12} = 8$ and $r_{43} = 10$. A fitted model predicts $\widehat{r}_{12} = 10$ and $\widehat{r}_{43} = 7$.

It also predicts $\widehat{r}_{13} = 4$, but r_{13} is unknown. What is the validation RMSE?

A. $13/2$

B. $\sqrt{13/2}$

C. $\sqrt{29/3}$

D. $\sqrt{13/3}$

Quiz Q2 answer. The correct answer is B.

Only the two validation pairs have known target ratings:

$$\mathrm{RMSE}_{\mathrm{val}} \;=\; \sqrt{\frac{(8-10)^2+(10-7)^2}{2}} \;=\; \sqrt{\frac{13}{2}} \;\approx\; 2.55.$$

The unknown rating r_{13} contributes nothing to this evaluation.

A prediction alone is not a known target value.